SESSION XIV

(Trivedi) Goodness of fit

We have assumed the distribution function of X is known.

Problems studied so far $\begin{cases} \text{Estimation of parameters} \\ \text{Hypothesis testing concerning parameters} \end{cases}$

New Problem Given experimental observations. Postulate distribution of X.

Question: How to test goodness of fit?

Methods — Graphical
Analytical.

Test — Chi Square (χ^2) Test — Versatile test

Method to test — whether a given population characterized by a particular r.v

Exponential, Gamma Normal

Chi-Square Distribution

square of $Z \sim N(0,1)$ is chi-square with 1 d.f. ie $Z^2 \sim \chi^2_1$.

In general n ind. rvs — $X_i \sim N(\mu_i, \sigma_i^2)$ $i = 1, \dots n$; Independent RVs.

$$Z_i = \frac{X_i - \mu_i}{\sigma_i^2}, \quad i = 1, \dots n.$$

$$\chi^2 = \sum_{i=1}^{n} \left(\frac{x_i - \mu_i}{\sigma_i} \right)^2 = \sum_{i=1}^{n} Z_i^2 \quad \text{is}$$

Chi-square with n d.f.

Mgf $\quad M_{\chi^2}(t) = (1-2t)^{-n/2}$ for $t < \frac{1}{2}$.

Recall If $X \sim \text{Gamma}(\frac{1}{2}, \frac{n}{2})$, then it is chi-square distribution with n degrees of freedom.

Note: Random sampling from normal population with mean μ and variance σ^2.

$$\bar{x} = \frac{\sum x_i}{n}, \quad \text{var}\,\bar{x} = \frac{\sigma^2}{n}$$

Therefore $\quad \dfrac{\bar{x} - \mu}{\sigma/\sqrt{n}} \sim N(0,1) \Rightarrow \left(\dfrac{\bar{x} - \mu}{\sigma/\sqrt{n}} \right)^2 \sim \chi^2_1$.

<u>Important Result</u>

In a <u>large</u> random sample with n_i observed frequency in ith class ~~with~~ and np_i the corresponding expected frequency, then

$$\sum_{i=1}^{k} \left[\frac{(n_i - np_i)^2}{np_i} \right] \sim \chi^2_{k-1}$$

where $\displaystyle\sum_{i=1}^{k} n_i = n$, $(i = 1, \dots k)$

<u>Multinomial Distribution</u>

Consider a sample of size n with observed value of a multinomial random vector

with parameters p_i, $i = 1, 2, \ldots k$. Then

$$P[X_1 = x_1, X_2 = x_2, \ldots, X_k = n_k]$$

$$= \frac{n!}{x_1!\, x_2! \cdots x_k!} p_1^{x_1} p_2^{x_2} \cdots p_k^{x_k} ; \quad n = \sum_{i=1}^{k} x_i$$

$$\sum p_i = 1.$$

The marginal pmf of X_i

$$p_{X_i}(n_i) = \sum_{n : \sum_{j \neq i} n_j = n - n_i} \frac{(n-n_i)!\, p_1^{n_1} \cdots p_{i-1}^{n_{i-1}} p_{i+1}^{n_{i+1}} \cdots p_k^{n_k}}{n_1! \cdots n_{i-1}!\, n_{i+1}! \cdots n_k!}$$

$$= \binom{n}{n_i} p_i^{n_i} (p_1 + p_2 \cdots + p_{i-1} + p_{i+1} \cdots + p_k)^{n-n_i}$$

$$= \binom{n}{n_i} p_i^{n_i} (1-p_i)^{n-n_i} \sim \text{Binomial pmf.}$$

(See Trivedi 2nd Ed p102).

It can be shown for large n such that n_i $(i = 1, \ldots k)$ are not small then using Stirling's formula, the random variable

$$U = \sum_{i=1}^{k} \frac{(n_i - np_i)^2}{np_i} \sim \chi^2_{k-1}.$$

The above result can be stated as

Thm: Let $X_1, X_2, \ldots X_n$ be a multinomial random variable with parameters $n, p_1, \ldots p_{k-1}$, then the random variable

$$U_k = \sum_{i=1}^{k} \frac{(x_i - np_i)^2}{np_i}$$

is asymptotically distributed as χ^2 rv with $k-1$ df.

Ex. A die is rolled 120 times, the following information was obtained.

Result	1	2	3	4	5	6
Frequency	20	30	20	25	15	10

Test the hypothesis that the die is fair at level $\alpha = 0.05$.

Soln

$H_0 : p_i = \frac{1}{6} \quad i = 1, 2 \ldots 6$

H_1 : not H_0

We reject H_0 if

$$u = \sum_{i=1}^{6} \frac{\left(x_i - 120 \times \frac{1}{6}\right)^2}{120 \times \frac{1}{6}} > \chi^2_{5, 0.05}$$

We find

$$u = 0 + \frac{10^2}{20} + 0 + \frac{5^2}{20} + \frac{10^2}{20} = 12.5$$

Since

$$\chi^2_{5, 0.05} = 11.07$$

We reject H_0 at 5% level of significance

However, at $\alpha = 0.025$, then

$$\chi^2_{5,0.025} = 12.8$$

Since $\chi^2_{5,0.025}$ is more than the ~~extract~~ calculated value, we cannot reject H_0.

χ^2 statistic

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i} \sim \chi^2_m ,$$

$m = k-1$. Subtract one from each independent parameter that is estimated from the data to generate expected E_i, $i = 1, \ldots k$.

Note: In the derivation of test statistic multivariate normal approximation to multinomial distribution is employed.

1. Each np_i value should be moderately large
   — rule of thumb $np_i \geq 5$.

2. Combine categories to form a single category if expected number in several categories are small

3. If r parameters are to be estimated, the χ^2-statistic loses r d.f, then χ^2 is distributed with $k-r-1$.

Ex. The number of errors discovered in a system program is Poisson distributed. The number of errors in each week period is given in Table. The total number of errors in 50 weeks was 95. Test the null hypothesis of Poisson distribution at 5% level of significance

No of errors (i) in one week period	No of one-week periods with i errors	Poisson prob.	Expected frequencies
0	14	0.150	7.50
1	11	0.284	14.20
2	9	0.270	13.50
3	6	0.171	8.55
4	5	0.081	4.05
5+	5	0.044	2.20

Soln: To compute Poisson probabilities

$$\hat{\lambda} = \frac{\text{total no. of observed errors}}{\text{total no. of weeks observed}} = \frac{95}{50} = 1.9/\text{week}$$

$$\chi^2 = \frac{(14-7.50)^2}{7.50} + \cdots + \frac{(5-2.20)^2}{2.20} = 12.401 \text{ (Calculated)}$$

$$d.f = 6-1-1 = 4$$

$$\chi^2_{4,0.05} = 9.488 \text{ (Tabulated)}$$

H_0 is rejected as calculated value exceeds the critical value.

Ex. Show that if $p_i^0 = \frac{1}{k}$, $i = 1, \dots k$ then the statistic

$$Q = \sum_{i=1}^{k} \frac{(N_i - n p_i^0)^2}{n p_i^0},$$

can be written in the form

$$Q = \left(\frac{k}{n} \sum_{i=1}^{k} N_i^2 \right) - n.$$

Soln

$$Q = \sum_{i=1}^{k} \frac{\left(N_i - \frac{n}{k}\right)^2}{n/k} = \frac{k}{n} \sum_{i=1}^{k} \left(N_i^2 - \frac{2n}{k} N_i + \frac{n^2}{k^2} \right)$$

$$= \frac{k}{n} \sum N_i^2 - \frac{k}{n} \left(+ \frac{2n}{k} \sum N_i - \frac{n^2}{k} \right)$$

$$= \frac{k}{n} \sum N_i^2 - \frac{k}{n} \left(+ \frac{2n^2}{k} - \frac{n^2}{k} \right) = \frac{k}{n} \sum N_i^2 - n$$

Ex Suppose that in a sequence of Bernoulli trials, the probability p of success on each trial is unknown. Further, p_0 is a given number in the interval $(0,1)$ and it is desired to test the following hypothesis.

$$H_0 : p = p_0$$

$$H_1 : p \neq p_0$$

Let $\bar{X}_n$ be the proportion of success in n trials. Show that the statistic

$$Q = \left[n (\bar{X}_n - p_0)^2 \right] / \left[p_0 (1 - p_0) \right].$$

Soln

$$Q = \sum_{i=1}^{k} \frac{(N_i - np_i^0)^2}{np_i^0}$$

$\bar{X}_n$ denotes proportion of successes

$n\bar{X}_n$ . . . . the number of successes

Therefore $n(1-\bar{X}_n)$ . . . the number of failures

$$Q = \sum_{i=1}^{2} \frac{(N_i - np_i^0)^2}{np_i^0} = \frac{(n\bar{X}_n - np_0)^2}{np_0} + \frac{[n(1-\bar{X}_n) - n(1-p_0)]^2}{n(1-p_0)}$$

$$= n(\bar{X}_n - p_0)^2 \left(\frac{1}{p_0} + \frac{1}{1-p_0}\right)$$

$$= \frac{n(\bar{X}_n - p_0)^2}{p_0(1-p_0)}$$

Example: Simulate 200 pseudo random numbers between 0 and 1 and divide the unit interval into $k = 10$ intervals of length 0.1. Investigate the randomness of the generator by using χ^2-test of the hypothesis that each 10 intervals has the same probability of containing a pseudo random number.

i	0	1	2	3	4	5	6	7	8	9
N_i	25	16	19	20	20	22	24	15	14	25

Soln:

Here $p_i^0 = \frac{1}{10}$, $i = 1, \ldots 10$

$$Q = \sum_{i=1}^{10} \frac{(N_i - 20)^2}{20} = 7.4$$

$$\chi^2_{9, 0.05} =$$

Ex. Suppose that six rvs $X_1, X_2, \ldots X_6$ from a random sample from standard normal distribution, and let

$$Y = (X_1 + X_2 + X_3)^2 + (X_4 + X_5 + X_6)^2$$

Determine c such that cY has a χ^2 distn.

Soln

Let

$Y_1 = X_1 + X_2 + X_3$, $Y_2 = X_4 + X_5 + X_6$

$X_i \sim N(0, 1)$ $i = 1, \ldots 6$

X_is are independent

$E[Y_1] = 0$ $V[Y_1] = 3$

$E[Y_2] = 0$ $V(Y_2) = 3$

If $c = \frac{1}{\sqrt{3}}$, then $E[Y_i] = 0$, $V[Y_i] = 1$, $i = 1, 2$

and $Y_i \sim N(0, 1)$; $i = 1, 2$

$Y_1^2 \sim \chi^2_1$ and $Y_2^2 \sim \chi^2_1$

$Y = Y_1^2 + Y_2^2$

Therefore $Y \sim \chi^2_2$.

Ex. The following data were obtained from a table of random numbers of normal distribution with 0 and variance 1.

0.464	0.137	2.455	−0.323	−0.068
0.906	−0.513	−0.525	0.595	0.881
−0.482	1.678	−0.057	−1.229	−0.486
−1.787	−0.261	1.237	1.046	−1.508

Test the hypothesis that the distribution function F from which the data came is normal with 0 and variance 1.

Soln

The distributional form is specified

Let us choose three intervals Y_1, Y_2 and Y_3 viz

$(-\infty, -0.5]$, $(-0.5, 0.5]$, $(0.5, \infty)$.

Frequencies of $Y_1 = 5$, $Y_2 = 8$, and $Y_3 = 7$

$Z \sim N(0,1)$, the corresponding probabilities of

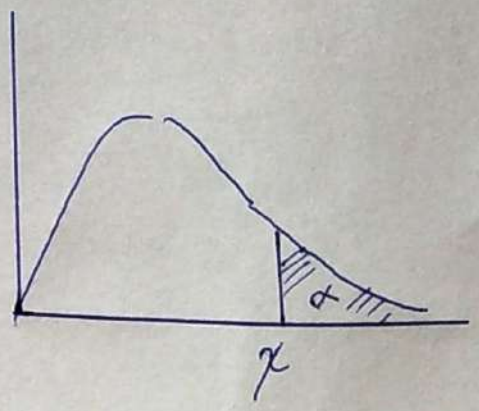
$p_1 = P(Y_1) = 0.3085$, $p_2 = 0.3830$, $p_3 = 0.3085$

$$Q = \sum_{i=1}^{3} \frac{(y_i - np_i)^2}{np_i} = \frac{(5 - 20 \times .3085)^2}{6.17} + \cdots + \cdots$$

$$= \cdots < 1$$

Also $\chi^2_{2,0.05} = 5.99$.

We cannot reject at 5% level.



Critical value of χ^2 distribution

χ^2 - Goodness of fit

1. Count O_i observed elements in Category C_i $(i=1,\dots k)$

2. Calculate E_i (on the basis of H_0)

3. Calculate χ^2 statistic

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

4 Calculate no. of d.f

5 Find $\chi^2_{m,\alpha}$ such that probability that a chi-square r.v. with mdf will exceed $\chi^2_{m,\alpha}$ is α.

6 If $\chi^2 \geq \chi^2_{m,\alpha}$ reject H_0; otherwise accept H_0

(From Arnold A Allen)